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Experiment-10

1. Algorithm

Step 1: Start.

Step 2: Input a positive integer `n`.

Step 3: Display the menu:

1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares

Step 4: Input the user's choice.

Step 5: Use `switch(choice)` to perform the selected operation.

Step 6: If choice is 1:

- Set `sum = 0`.
- Copy `n` into `temp`.
- Using a **while loop**, take each digit using `temp % 10`.
- Find the square of the digit and add it to `sum`.
- Remove the last digit using `temp = temp / 10`.
- Display the sum of squares.

Step 7: If choice is 2:

- Set `sum = 0`.
- Using a **for loop**, take each digit of the number.
- Find the cube of each digit and add it to `sum`.
- Display the sum of cubes.

Step 8: If choice is 3:

- Set `squareSum = 0` and `cubeSum = 0`.
- Copy `n` into `temp`.

- Using a **do-while loop**, take each digit.
- Add the square of the digit to `squareSum`.
- Add the cube of the digit to `cubeSum`.
- Remove the last digit.
- Calculate `difference = cubeSum - squareSum`.
- Display the difference.

Step 9: If the user enters any other choice, display "Invalid choice".

Step 10: Stop.

2. Test Case Table

Test Case	Input	Expected Outputs	Actual Output	Results
1	Number = 123, Choice = 1	Sum of squares = 14	Sum of squares = 14	Pass
2	Number = 123, Choice = 2	Sum of cubes = 36	Sum of cubes = 36	Pass
3	Number = 123, Choice = 3	Difference = 22	Difference = 22	Pass
4	Number = 45, Choice = 1	Sum of squares = 41	Sum of squares = 41	Pass
5	Number = 45, Choice = 2	Sum of cubes = 189	Sum of cubes = 189	Pass
6	Number = 45, Choice = 3	Difference = 148	Difference = 148	Pa

3. C Program

```
#include <stdio.h>
int main()
{
    int n, temp, digit, choice;
    int sum, squareSum, cubeSum, difference;
    printf("Enter a positive integer: ");
    scanf("%d", &n);
    printf("\n1. Sum of squares of digits");
    printf("\n2. Sum of cubes of digits");
    printf("\n3. Difference between sum of cubes and sum of squares");
    printf("\nEnter your choice: ");
    scanf("%d", &choice);
    switch(choice)
    {
```

```

case 1:
    temp = n;
    sum = 0;
    while(temp > 0)
    {
        digit = temp % 10;
        sum = sum + digit * digit;
        temp = temp / 10;
    }
    printf("Sum of squares = %d\n", sum);
    break;
case 2:
    sum = 0;
    for(temp = n; temp > 0; temp = temp / 10)
    {
        digit = temp % 10;
        sum = sum + digit * digit * digit;
    }
    printf("Sum of cubes = %d\n", sum);
    break;
case 3:
    temp = n;
    squareSum = 0;
    cubeSum = 0;
    do
    {
        digit = temp % 10;
        squareSum = squareSum + digit * digit;
        cubeSum = cubeSum + digit * digit * digit;
        temp = temp / 10;
    }
    while(temp > 0);
    difference = cubeSum - squareSum;

    printf("Difference = %d\n", difference);
    break;
default:
    printf("Invalid choice\n");
}
return 0;
}

```

4.Compilation and Execution

```
(kali㉿kali)-[~]  
$ gcc hdoc11.c -o hdoc11  
  
(kali㉿kali)-[~]  
$ ./hdoc11  
Enter a positive integer: 123  
  
1. Sum of squares of digits  
2. Sum of cubes of digits  
3. Difference between sum of cubes and sum of squares  
Enter your choice: 1  
Sum of squares = 14
```

```
(kali㉿kali)-[~]  
$ gcc hdoc11.c -o hdoc11  
  
(kali㉿kali)-[~]  
$ ./hdoc11  
Enter a positive integer: 123  
  
1. Sum of squares of digits  
2. Sum of cubes of digits
```

```
3. Difference between sum of cubes and sum of squares  
Enter your choice: 2  
Sum of cubes = 36
```

```
(kali㉿kali)-[~]  
$ gcc hdoc11.c -o hdoc11  
  
(kali㉿kali)-[~]  
$ ./hdoc11  
Enter a positive integer: 123  
  
1. Sum of squares of digits  
2. Sum of cubes of digits  
3. Difference between sum of cubes and sum of squares  
Enter your choice: 3  
Difference = 22
```

```
(kali㉿kali)-[~]  
$ gcc hdoc11.c -o hdoc11  
  
(kali㉿kali)-[~]  
$ ./hdoc11
```

```
(kali㉿kali)-[~]  
$ gcc hdoc11.c -o hdoc11  
  
(kali㉿kali)-[~]  
$ ./hdoc11  
Enter a positive integer: 45  
  
1. Sum of squares of digits  
2. Sum of cubes of digits  
3. Difference between sum of cubes and sum of squares  
Enter your choice: 1  
Sum of squares = 41
```

```
(kali㉿kali)-[~]  
$ gcc hdoc11.c -o hdoc11  
  
(kali㉿kali)-[~]  
$ ./hdoc11  
Enter a positive integer: 45  
  
1. Sum of squares of digits  
2. Sum of cubes of digits  
3. Difference between sum of cubes and sum of squares  
Enter your choice: 2  
Sum of cubes = 189
```

```
(kali㉿kali)-[~]  
$ gcc hdoc11.c -o hdoc11
```

```
(kali㉿kali)-[~]  
$ ./hdoc11  
Enter a positive integer: 45  
  
1. Sum of squares of digits  
2. Sum of cubes of digits  
3. Difference between sum of cubes and sum of squares  
Enter your choice: 3  
Difference = 148
```